Lecture on Conformal Prediction (2/4)

Margaux Zaffran

October 29-30, 2026

LIDAM – ISBA seminar series

FONDATION MATHÉMATIQUE JACQUES HADAMARD

Laboratoire de Mathématiques d'Orsay

LIDAM | Louvain Institute of Data Analysis and Modeling in economics and statistics

ISBA

This morning's lecture

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory)

Achievable definition of distribution-free validity

$\widehat{C}_\alpha$ = estimated predictive set based on n data points.

Definition (Distribution-free validity)**.**

$\widehat{C}_\alpha$ achieves distribution-free validity if:

- for any distribution $\mathcal{D}$,
- for any associated exchangeable joint distribution $\mathcal{D}^{\text{exch}(n+1)}$,

we have that:

$$\mathbb{P}_{\mathcal{D}^{\text{exch}(n+1)}}\left(Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right)\right) \geq 1-\alpha.$$

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal} + 1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

Ex 2: $\mathsf{s}(\hat{A}(X_i), Y_i) := \max\left(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right)$ in CQR

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

Ex 2: $\mathsf{s}(\hat{A}(X_i), Y_i) := \max\left(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right)$ in CQR

4. Compute the $1-\alpha$ quantile of these scores, noted $q_{1-\alpha}(\mathcal{S})$

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

Ex 2: $\mathsf{s}(\hat{A}(X_i), Y_i) := \max\left(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right)$ in CQR

4. Compute the $1-\alpha$ quantile of these scores, noted $q_{1-\alpha}(\mathcal{S})$
5. For a new point X_{n+1}, return

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

Ex 2: $\mathsf{s}(\hat{A}(X_i), Y_i) := \max\left(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right)$ in CQR

4. Compute the $1-\alpha$ quantile of these scores, noted $q_{1-\alpha}(\mathcal{S})$
5. For a new point X_{n+1}, return

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

Ex 1: $\widehat{C}_\alpha(X_{n+1}) = [\hat{\mu}(X_{n+1}) \pm q_{1-\alpha}(\mathcal{S})]$

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathrm{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathrm{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

Ex 2: $\mathrm{s}(\hat{A}(X_i), Y_i) := \max\left(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right)$ in CQR

4. Compute the $1-\alpha$ quantile of these scores, noted $q_{1-\alpha}(\mathcal{S})$
5. For a new point X_{n+1}, return

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

Ex 2: $\widehat{C}_\alpha(X_{n+1}) = [\widehat{QR}_{\mathrm{lower}}(X_{n+1}) - q_{1-\alpha}(\mathcal{S});$

$$\widehat{QR}_{\mathrm{upper}}(X_{n+1}) + q_{1-\alpha}(\mathcal{S})]$$

SCP is defined by the conformity score function



1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ *by training the algorithm* $\mathcal{A}$ *on the proper training set*
3. On the calibration set, obtain $\#\mathrm{Cal}+1$ conformity scores

$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

Ex 2: $\mathsf{s}(\hat{A}(X_i), Y_i) := \max\left(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right)$ in CQR

4. Compute the $1-\alpha$ quantile of these scores, noted $q_{1-\alpha}(\mathcal{S})$
5. For a new point X_{n+1}, return

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

$\hookrightarrow$ The definition of the conformity scores is crucial, as they incorporate almost all the information: data + underlying model

Split Conformal Prediction: summary

- **Simple** procedure which quantifies the uncertainty of **any** predictive model $\hat{A}$ by returning predictive regions
- **Finite-sample** guarantees
- **Distribution-free** as long as the data are exchangeable (and so are the scores)

Split Conformal Prediction: summary

- **Simple** procedure which quantifies the uncertainty of **any** predictive model $\hat{A}$ by returning predictive regions
- **Finite-sample** guarantees
- **Distribution-free** as long as the data are exchangeable (and so are the scores)
- **Marginal** theoretical guarantee over the joint (X, Y) distribution, and not conditional, i.e., no guarantee that for any x:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right) \cancel{| X_{n+1} = x}\right\} \geq 1 - \alpha.$$

Split Conformal Prediction: summary

- **Simple** procedure which quantifies the uncertainty of **any** predictive model $\hat{A}$ by returning predictive regions
- **Finite-sample** guarantees
- **Distribution-free** as long as the data are exchangeable (and so are the scores)
- **Marginal** theoretical guarantee over the joint (X, Y) distribution, and not conditional, i.e., no guarantee that for any x:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right) \cancel{\left| X_{n+1} = x \right.}\right\} \geq 1 - \alpha.$$

  $\hookrightarrow$ marginal also over the whole calibration set and the test point!

On the design choices of conformity scores and (empirical) conditional guarantees

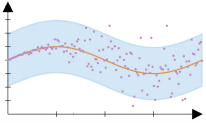
- Impact of the calibration set on the coverage
- On distribution-free X-conditional validity
- Y-conditional validity
- Case study: healthcare

SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}\left(\mathcal{S}\right)\}$$

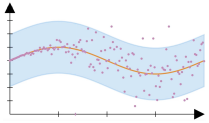
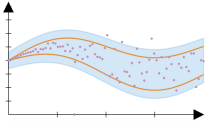
SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}\left(\mathcal{S}\right)\}$$

	Standard SCP Vovk et al. (2005)		
$\mathsf{s}\left(\hat{A}(X), Y\right)$	$\lvert\hat{\mu}(X) - Y\rvert$		
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}\left(\mathcal{S}\right)]$		
Visu.			
✓	black-box around a "usable" prediction		
✗	not adaptive		

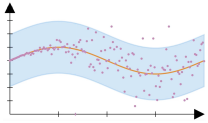
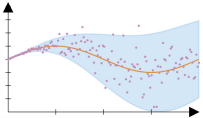
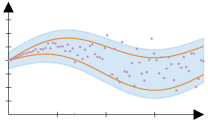
SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

	Standard SCP Vovk et al. (2005)		**CQR** Romano et al. (2019)
$\mathrm{s}(\hat{A}(X), Y)$	$\lvert\hat{\mu}(X) - Y\rvert$		$\max(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X) - Y,$ $Y - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X))$
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$		$[\widehat{\mathrm{QR}}_{\mathrm{lower}}(x) - q_{1-\alpha}(\mathcal{S});$ $\widehat{\mathrm{QR}}_{\mathrm{upper}}(x) + q_{1-\alpha}(\mathcal{S})]$
Visu.			
✓	black-box around a "usable" prediction		adaptive
✗	not adaptive		no black-box around a "usable" prediction

SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

	Standard SCP Vovk et al. (2005)	**Locally weighted SCP** Lei et al. (2018)	**CQR** Romano et al. (2019)
$\mathrm{s}(\hat{A}(X), Y)$	$\lvert\hat{\mu}(X) - Y\rvert$	$\dfrac{\lvert\hat{\mu}(X) - Y\rvert}{\hat{\rho}(X)}$	$\max(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X) - Y,$ $Y - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X))$
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})\hat{\rho}(x)]$	$[\widehat{\mathrm{QR}}_{\mathrm{lower}}(x) - q_{1-\alpha}(\mathcal{S});$ $\widehat{\mathrm{QR}}_{\mathrm{upper}}(x) + q_{1-\alpha}(\mathcal{S})]$
Visu.			
✓	black-box around a "usable" prediction	black-box around a "usable" prediction	adaptive
✗	not adaptive	limited adaptiveness	no black-box around a "usable" prediction

Challenges that we are going to explore in this afternoon's lecture

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory)

Challenges that we are going to explore in this afternoon's lecture

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)

Challenges that we are going to explore in this afternoon's lecture

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)
3. Towards conditional coverage? (practical session)

Challenges that we are going to explore in this afternoon's lecture

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)
3. Towards conditional coverage? (practical session) (theory) (case studies)

On the design choices of conformity scores and (empirical) conditional guarantees

Impact of the calibration set on the coverage

On distribution-free X-conditional validity

Y-conditional validity

Case study: healthcare

Calibration-condition coverage distribution under no tie

Theorem (Distribution conditional on the calibration data).

If the scores are a.s. distinct, SCP outputs $\widehat{C}_\alpha$ such that for any distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$:

$$\mathbb{P}_{\mathcal{D}}\left(Y_{n+1} \in \widehat{C}_\alpha(X_{n+1}) | (X_i, Y_i)_{i \in \mathrm{Cal}}\right) \sim \beta\left(k_\alpha, \#\mathrm{Cal} + 1 - k_\alpha\right),$$

with $k_\alpha = \lceil (1-\alpha)(\#\mathrm{Cal}+1) \rceil$.

From the β distribution, we get that it has

expectation $\dfrac{k_\alpha}{k_\alpha + \#\mathrm{Cal} + 1 - k_\alpha} = \dfrac{k_\alpha}{\#\mathrm{Cal}+1} = \dfrac{\lceil (1-\alpha)(\#\mathrm{Cal}+1) \rceil}{\#\mathrm{Cal}+1} \geq 1-\alpha,$

and variance $\dfrac{k_\alpha\left(\#\mathrm{Cal} + 1 - k_\alpha\right)}{\left(\#\mathrm{Cal}+1\right)^2\left(\#\mathrm{Cal}+2\right)} \approx \dfrac{\alpha(1-\alpha)}{\#\mathrm{Cal}+2}.$

Probably Approximately Correct bounds on calibration-conditional coverage (Vovk, 2012; Bian and Barber, 2023)

Theorem (calibration conditional validity of SCP).

SCP outputs $\widehat{C}_\alpha$ such that for any distribution $\mathcal{D}$ and any $0 < \delta \leq 0.5$:

$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\mathbb{P}_{\mathcal{D}}\left(Y_{n+1} \notin \widehat{C}_{n,\alpha}\left(X_{n+1}\right) \,|\, (X_i, Y_i)_{i=1}^n\right) \leq \alpha + \sqrt{\frac{\log(1/\delta)}{2\#\mathrm{Cal}}}\right) \geq 1-\delta.$$

$\hookrightarrow$ controls the deviation of miscoverage with respect to the nominal level of a predictive set built on a given calibration set.

On the design choices of conformity scores and (empirical) conditional guarantees

- Impact of the calibration set on the coverage
- On distribution-free X-conditional validity
- Y-conditional validity
- Case study: healthcare

Definition of distribution-free features conditional validity

$\widehat{C}_\alpha$ = estimated predictive set based on n data points.

Definition (Distribution-free X-conditional validity)**.**

$\widehat{C}_\alpha$ achieves distribution-free X-conditional validity if:

- for any distribution $\mathcal{D}$,
- for any associated exchangeable joint distribution $\mathcal{D}^{\mathrm{exch}(n+1)}$,

we have that:

$$\mathbb{P}_{\mathcal{D}^{\mathrm{exch}(n+1)}}\left(Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right) | X_{n+1}\right) \overset{a.s.}{\geq} 1-\alpha.$$

Informative conditional coverage as such is impossible

Theorem (Impossibility results Vovk (2012); Lei and Wasserman (2014)).

If $\widehat{C}_\alpha$ is distribution-free X-conditionally valid, then, for any $\mathcal{D}$, for $\mathcal{D}_X$–almost all $\mathcal{D}_X$–non-atoms $x \in \mathcal{X}$, it holds:

- *Regression:* $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(\text{mes}\left(\widehat{C}_\alpha(x)\right) = \infty\right) \geq 1 - \alpha$,
- *Classification:* for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(y \in \widehat{C}_\alpha(x)\right) \geq 1 - \alpha$.

Informative conditional coverage as such is impossible

Theorem (Impossibility results Vovk (2012); Lei and Wasserman (2014)).

If $\widehat{C}_\alpha$ is distribution-free X-conditionally valid, then, for any $\mathcal{D}$, for $\mathcal{D}_X$–almost all $\mathcal{D}_X$–non-atoms $x \in \mathcal{X}$, it holds:

- *Regression:* $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(\text{mes}\left(\widehat{C}_\alpha(x)\right) = \infty\right) \geq 1 - \alpha$,
- *Classification:* for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(y \in \widehat{C}_\alpha(x)\right) \geq 1 - \alpha$.

$\hookrightarrow$ distribution-free X-conditional hardness result apply beyond CP

Informative conditional coverage as such is impossible

Theorem (Impossibility results Vovk (2012); Lei and Wasserman (2014))**.**

If $\widehat{C}_\alpha$ is distribution-free X-conditionally valid, then, **for any** $\mathcal{D}$, for $\mathcal{D}_X$–almost all $\mathcal{D}_X$–non-atoms $x \in \mathcal{X}$, it holds:

- *Regression:* $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(\text{mes}\left(\widehat{C}_\alpha(x)\right) = \infty\right) \geq 1 - \alpha$,
- *Classification:* for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(y \in \widehat{C}_\alpha(x)\right) \geq 1 - \alpha$.

$\hookrightarrow$ distribution-free X-conditional hardness result apply beyond CP

$\hookrightarrow$ X-conditional estimators are overly large even on easy cases

Informative conditional coverage as such is impossible

Theorem (Impossibility results Vovk (2012); Lei and Wasserman (2014)).

If $\widehat{C}_\alpha$ is distribution-free X-conditionally valid, then, for any $\mathcal{D}$, for $\mathcal{D}_X$–almost all $\mathcal{D}_X$–non-atoms $x \in \mathcal{X}$, it holds:

- *Regression:* $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(\text{mes}\left(\widehat{C}_\alpha(x)\right) = \infty\right) \geq 1-\alpha$,
- *Classification:* for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(y \in \widehat{C}_\alpha(x)\right) \geq 1-\alpha$.

$\hookrightarrow$ distribution-free X-conditional hardness result apply beyond CP

$\hookrightarrow$ X-conditional estimators are overly large even on easy cases

$\hookrightarrow$ the lower bounds are tight

Example (Naive estimator).

$\mathcal{C}_\alpha(\cdot;\xi) \equiv \mathcal{Y}\mathbb{1}\{\xi \leq 1-\alpha\} + \emptyset\mathbb{1}\{\xi > \alpha\}$, where $\xi \sim \mathcal{U}([0,1])$.

Informative conditional coverage as such is impossible

Theorem (Impossibility results Vovk (2012); Lei and Wasserman (2014)).

If $\widehat{C}_\alpha$ is distribution-free X-conditionally valid, then, for any $\mathcal{D}$, for $\mathcal{D}_X$–almost all $\mathcal{D}_X$–non-atoms $x \in \mathcal{X}$, it holds:

- *Regression:* $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(\text{mes}\left(\widehat{C}_\alpha(x)\right) = \infty\right) \geq 1 - \alpha$,
- **Classification:** for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n)}}\left(y \in \widehat{C}_\alpha(x)\right) \geq 1 - \alpha$.

$\hookrightarrow$ distribution-free X-conditional hardness result apply beyond CP

$\hookrightarrow$ X-conditional estimators are overly large even on easy cases

$\hookrightarrow$ the lower bounds are tight

$\hookrightarrow$ Classification: *every label* is likely to be included in $\widehat{C}_\alpha$.
$\widehat{C}_\alpha$ is likely to be large: for any $\mathcal{D}$, for $\mathcal{D}_X$–almost all $\mathcal{D}_X$–non-atoms $x \in \mathcal{X}$, $\mathbb{E}_{\mathcal{D}^{\otimes(n)}}\left[\#\widehat{C}_\alpha(x)\right] \geq (1-\alpha)\#\mathcal{Y}$.

Getting closer to X-conditional coverage

	X-conditionally valid	
"Non smooth" distribution		

Getting closer to X-conditional coverage

	X-conditionally valid	
"Non smooth" distribution	X-cov.: ✓ Length: ✓	

Getting closer to X-conditional coverage

	X-conditionally valid	
"Non smooth" distribution	X-cov.: ✓ Length: ✓	
"Smooth" distribution	X-cov.: ✓ Length: ✗	

Getting closer to X-conditional coverage

	X-conditionally valid	Non X-conditionally valid
"Non smooth" distribution	X-cov.: ✓ Length: ✓	X-cov.: ✗ Length: `not relevant`
"Smooth" distribution	X-cov.: ✓ Length: ✗	

Getting closer to X-conditional coverage

	X-conditionally valid	Non X-conditionally valid
"Non smooth" distribution	X-cov.: ✓ Length: ✓	X-cov.: ✗ Length: `not relevant`
"Smooth" distribution	X-cov.: ✓ Length: ✗	X-cov.: $\approx$ Length: ✓

Getting closer to X-conditional coverage

	X-conditionally valid	Non X-conditionally valid
"Non smooth" distribution	X-cov.: ✓ Length: ✓	X-cov.: ✗ Length: `not relevant`
"Smooth" distribution	X-cov.: ✓ Length: ✗	X-cov.: $\approx$ Length: ✓

- Asymptotic (with the sample size) conditional coverage
  $\hookrightarrow$ **Romano et al. (2019)**; Kivaranovic et al. (2020); Chernozhukov et al. (2021); Sesia and Romano (2021); Izbicki et al. (2022)

Non exhaustive references.

Getting closer to X-conditional coverage

	X-conditionally valid	Non X-conditionally valid
"Non smooth" distribution	X-cov.: ✓ Length: ✓	X-cov.: ✗ Length: `not relevant`
"Smooth" distribution	X-cov.: ✓ Length: ✗	X-cov.: $\approx$ Length: ✓

- Asymptotic (with the sample size) conditional coverage
  $\hookrightarrow$ Romano et al. (2019); Kivaranovic et al. (2020); Chernozhukov et al. (2021); Sesia and Romano (2021); Izbicki et al. (2022)
- Approximate conditional coverage
  $\hookrightarrow$ Romano et al. (2020); Guan (2022); Jung et al. (2023); Gibbs et al. (2023)
  Target $\mathbb{P}(Y^{(n+1)} \in \widehat{C}_\alpha\left(X^{(n+1)}\right) | X^{(n+1)} \in \mathcal{R}(x)) \geq 1 - \alpha$

Non exhaustive references.

Weaker notion of X-conditional validity (Barber et al., 2021)

Definition (distribution-free $(1-\alpha, \delta)$–X-conditional validity)**.**

Let $\delta > 0$ be a tolerance level.

An estimator $\widehat{C}_\alpha$ achieves distribution-free $(1-\alpha, \delta)$–X-conditional validity if for any distribution $\mathcal{D}$, for any $\mathcal{X} \subseteq \mathcal{X}$ such that $\mathbb{P}_{\mathcal{D}_X}(X \in \mathcal{X}) \geq \delta$, and for any associated exchangeable joint distribution $\mathcal{D}^{\text{exch}(n+1)}$, we have:

$$\mathbb{P}_{\mathcal{D}^{\text{exch}(n+1)}}\left(Y_{n+1} \in \widehat{C}_\alpha(X_{n+1}) \,|\, X_{n+1} \in \mathcal{X}\right) \geq 1-\alpha.$$

Weaker notion of X-conditional validity (Barber et al., 2021)

Definition (distribution-free $(1-\alpha, \delta)$–X-conditional validity)**.**

Let $\delta > 0$ be a tolerance level.
An estimator $\widehat{C}_\alpha$ achieves distribution-free $(1-\alpha, \delta)$–X-conditional validity if for any distribution $\mathcal{D}$, for any $\mathcal{X} \subseteq \mathcal{X}$ such that $\mathbb{P}_{\mathcal{D}_X}(X \in \mathcal{X}) \geq \delta$, and for any associated exchangeable joint distribution $\mathcal{D}^{\text{exch}(n+1)}$, we have:

$$\mathbb{P}_{\mathcal{D}^{\text{exch}(n+1)}}\left(Y_{n+1} \in \widehat{C}_\alpha(X_{n+1}) \,|\, X_{n+1} \in \mathcal{X}\right) \geq 1-\alpha.$$

Informal theorem (lower bound on $(1-\alpha, \delta)$–X-cond. valid efficiency)

An estimator achieving $(1-\alpha, \delta)$–X-conditional validity can not be more efficient than an estimator achieving **distribution-free marginal validity at the level** $1-\alpha\delta$.

$\hookrightarrow$ In practive, consider small $\delta \rightarrow$ unefficient predictive sets.

Pre-defined partition of the $\mathcal{X}$ space

$\widehat{C}_\alpha$ = estimated predictive set based on n data points.

$\mathcal{G}$ a set of "groups" (i.e., define G a random variable taking its values in $\mathcal{G}$).

Definition (Distribution-free $\mathcal{G}$-conditional validity ($\mathcal{G}$CV)).

$\widehat{C}_\alpha$ achieves distribution-free $\mathcal{G}$-conditional validity if for any distribution $\mathcal{D}$ on $(\mathcal{X}, \mathcal{G}, \mathcal{Y})$, it holds that:

$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(Y^{(n+1)} \in \widehat{C}_\alpha\left(X^{(n+1)}, G^{(n+1)}\right) | G^{(n+1)}\right) \overset{a.s.}{\geq} 1-\alpha.$$

Theorem (General $\mathcal{G}$CV hardness result)**.**

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$

Theorem (General $\mathcal{G}$CV hardness result).

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*

$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$$

- *Classification:*
  for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$

Theorem (General $\mathcal{G}$CV hardness result).

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$

Theorem (General $\mathcal{G}$CV hardness result).

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \text{ if } \underset{\mathcal{D}_G(g) \ll 1/n}{\simeq} 1-\alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right)$ if $\underset{\mathcal{D}_G(g) \ll 1/n}{\simeq} 1-\alpha$

Irreducible term: consider $\widehat{C}_\alpha$ outputting $\mathcal{Y}$ with probability $1-\alpha$ and $\emptyset$ otherwise.

Theorem (General $\mathcal{G}$CV hardness result)**.**

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1-\alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1-\alpha$

Irreducible term: consider $\widehat{C}_\alpha$ outputting $\mathcal{Y}$ with probability $1-\alpha$ and $\emptyset$ otherwise.

1. Similar result if we assume $G \perp\!\!\!\perp X$

Hardness results on G-conditional-validity (Zaffran et al., 2024)

Theorem (General $\mathcal{G}$CV hardness result)**.**

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$

Irreducible term: consider $\widehat{C}_\alpha$ outputting $\mathcal{Y}$ with probability $1 - \alpha$ and $\emptyset$ otherwise.

1. Similar result if we assume $G \perp\!\!\!\perp X$
2. Similar result if we assume $Y \perp\!\!\!\perp G \,|X$

Theorem (General $\mathcal{G}$CV hardness result).

If $\widehat{C}_\alpha$ is distribution-free $\mathcal{G}$-conditionally valid then for any distribution $\mathcal{D}$, for any group g such that $\mathcal{D}_G(g) := \mathbb{P}_{\mathcal{D}}(G = g) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, g\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, g\right)\right) \underset{\text{if } \mathcal{D}_G(g) \ll 1/n}{\simeq} 1 - \alpha$

Irreducible term: consider $\widehat{C}_\alpha$ outputting $\mathcal{Y}$ with probability $1 - \alpha$ and $\emptyset$ otherwise.

1. Similar result if we assume $G \perp\!\!\!\perp X$
2. Similar result if we assume $Y \perp\!\!\!\perp G \mid X$

$\Rightarrow$ need to restrict both the link between G and X, as well as between G and Y

Implications for $\mathcal{G}$CV in practice

	$\mathcal{G}$-conditionally valid even when $G \not\perp\!\!\!\perp (X, Y)$	
"Non smooth" distribution		

Implications for $\mathcal{G}$CV in practice

	$\mathcal{G}$-conditionally valid even when $G \not\perp\!\!\!\perp (X, Y)$	
"Non smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✓	

Implications for $\mathcal{G}$CV in practice

	$\mathcal{G}$-conditionally valid even when $G \not\perp\!\!\!\perp (X, Y)$	
"Non smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✓	
"Smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✗	

Implications for $\mathcal{G}$CV in practice

	$\mathcal{G}$-conditionally valid even when $G \not\perp\!\!\!\perp (X, Y)$	$\mathcal{G}$-conditionally valid at most if $G \perp\!\!\!\perp (X, Y)$
"Non smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✓	$\mathcal{G}$-cov.: ✗ Length: `not relevant`
"Smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✗	

Implications for $\mathcal{G}$CV in practice

	$\mathcal{G}$-conditionally valid even when $G \not\perp\!\!\!\perp (X, Y)$	$\mathcal{G}$-conditionally valid at most if $G \perp\!\!\!\perp (X, Y)$
"Non smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✓	$\mathcal{G}$-cov.: ✗ Length: `not relevant`
"Smooth" distribution	$\mathcal{G}$-cov.: ✓ Length: ✗	$\mathcal{G}$-cov.: $\approx$ Length: ✓

On the design choices of conformity scores and (empirical) conditional guarantees

- Impact of the calibration set on the coverage
- On distribution-free X-conditional validity
- Y-conditional validity
- Case study: healthcare

Achieving Y-conditional validity in classification

1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ (*by training $\mathcal{A}$ on the proper training set* $(X_i, Y_i)_{i \in \mathrm{Tr}}$)
3. **For** any candidate $y \in \mathcal{Y}$:
   On the calibration set, obtain a set of $\#\mathrm{Cal}_y + 1$ conformity scores :
   $$\mathcal{S}_y = \{S_i = \mathsf{s}\left(X_i, y; \hat{A}\right), i \in \mathrm{Cal} \text{ such that } Y_i = y\} \cup \{+\infty\}$$
4. For a new point X_{n+1}, return $\widehat{C}_{n,\alpha}(X_{n+1}) \left\{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S}_y)\right\}$

Achieving Y-conditional validity in classification

1. Randomly split the training data into a proper training set (size $\#\mathrm{Tr}$) and a calibration set (size $\#\mathrm{Cal}$)
2. Get $\hat{A}$ (*by training $\mathcal{A}$ on the proper training set* $(X_i, Y_i)_{i \in \mathrm{Tr}}$)
3. **For** any candidate $y \in \mathcal{Y}$:
   On the calibration set, obtain a set of $\#\mathrm{Cal}_y + 1$ conformity scores :
$$\mathcal{S}_y = \{S_i = \mathsf{s}\left(X_i, y; \hat{A}\right), i \in \mathrm{Cal} \text{ such that } Y_i = y\} \cup \{+\infty\}$$
4. For a new point X_{n+1}, return $\widehat{C}_{n,\alpha}(X_{n+1}) \left\{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S}_y)\right\}$

$\hookrightarrow$ What if there is a high class imbalance?

Ding et al. (2023) proposed to instead obtain **cluster**-conditional coverage, see further developments in Ding et al. (2026).

On the design choices of conformity scores and (empirical) conditional guarantees

- Impact of the calibration set on the coverage
- On distribution-free X-conditional validity
- Y-conditional validity
- Case study: healthcare

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

- Medical application
- Image based task
- Pixel by pixel analysis $\rightsquigarrow$ applications to segmentation for self-driving cars

Image-to-Image Regression with Distribution-Free Uncertainty Quantification and Applications in Imaging

Anastasios N. Angelopoulos [*1] Amit Kohli [*1] Stephen Bates [1] Michael I. Jordan [1] Jitendra Malik [1]
Thayer Alshaabi [2] Srigokul Upadhyayula [2 3] Yaniv Romano [4]

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

- Medical application
- Image based task
- Pixel by pixel analysis $\rightsquigarrow$ applications to segmentation for self-driving cars

1. **Task**: *Image to Image regression* – for each pixel of an image, predict a real valued output from the entire image.
2. **UQ Goal:** provide a predictive interval for each pixel, such that the output is in the interval at least 90% of the time.

Image-to-Image Regression with Distribution-Free Uncertainty Quantification and Applications in Imaging

Anastasios N. Angelopoulos [*1] Amit Kohli [*1] Stephen Bates [1] Michael I. Jordan [1] Jitendra Malik [1]
Thayer Alshaabi [2] Srigokul Upadhyayula [2 3] Yaniv Romano [4]

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

- Medical application
- Image based task
- Pixel by pixel analysis $\rightsquigarrow$ applications to segmentation for self-driving cars

1. **Task**: *Image to Image regression* – for each pixel of an image, predict a real valued output from the entire image.
2. **UQ Goal:** provide a predictive interval for each pixel, such that the output is in the interval at least 90% of the time.

Image-to-Image Regression with Distribution-Free Uncertainty Quantification and Applications in Imaging

Anastasios N. Angelopoulos [*1] Amit Kohli [*1] Stephen Bates [1] Michael I. Jordan [1] Jitendra Malik [1]
Thayer Alshaabi [2] Srigokul Upadhyayula [2 3] Yaniv Romano [4]

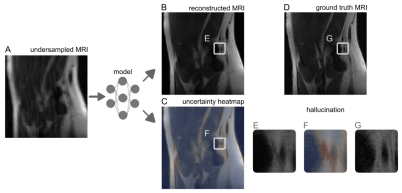


Figure 1. **An algorithmic MRI reconstruction with uncertainty.** A rapidly acquired but undersampled MR image of a knee (A) is fed into a model that predicts a sharp reconstruction (B) with calibrated uncertainty (C). In (C), red means high uncertainty and blue means low uncertainty. Wherever the reconstruction contains hallucinations, the uncertainty is high; see the hallucination in the image patch (E), which has high uncertainty in (F), and does not exist in the ground truth (G). For experimental details, see Section 3.4.

Figure 1: Image from Angelopoulos et al. (2022)

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Method:

1. Split conformal prediction method - isolate calibration set
2. On the proper training set, learn:
   - Mean regressor - $\hat{\mu} : \mathbb{R}^{NM} \rightarrow [0; 1]$

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Method:

1. Split conformal prediction method - isolate calibration set
2. On the proper training set, learn:
   - Mean regressor - $\hat{\mu} : \mathbb{R}^{NM} \to [0; 1]$
   - Heuristic notion of uncertainty: $\tilde{u}, \tilde{\ell} : \mathbb{R}^{NM} \to [0; 1]$, such that

$$[\hat{\mu}(X) - \tilde{\ell}(X); \hat{\mu}(X) + \tilde{u}(X)]$$

   $\to$ 3 regressors are used

   4 techniques are experimented for these regressors, including QR.

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Method:

1. Split conformal prediction method - isolate calibration set
2. On the proper training set, learn:
   - Mean regressor - $\hat{\mu} : \mathbb{R}^{NM} \to [0; 1]$
   - Heuristic notion of uncertainty: $\tilde{u}, \tilde{\ell} : \mathbb{R}^{NM} \to [0; 1]$, such that

     $$[\hat{\mu}(X) - \tilde{\ell}(X); \hat{\mu}(X) + \tilde{u}(X)]$$

     $\to$ 3 regressors are used

     4 techniques are experimented for these regressors, including QR.
3. Calibration step: leverage the calibration set.
   - In spirit, almost equivalent to CQR but with a multiplicative form.
   - Precisely, relies on RCPS (Bates et al., 2021)

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Method:

1. Split conformal prediction method - isolate calibration set
2. On the proper training set, learn:
   - Mean regressor - $\hat{\mu} : \mathbb{R}^{NM} \to [0; 1]$
   - Heuristic notion of uncertainty: $\tilde{u}, \tilde{\ell} : \mathbb{R}^{NM} \to [0; 1]$, such that

$$[\hat{\mu}(X) - \tilde{\ell}(X); \hat{\mu}(X) + \tilde{u}(X)]$$

   $\to$ 3 regressors are used

   4 techniques are experimented for these regressors, including QR.
3. Calibration step: leverage the calibration set.
   - In spirit, almost equivalent to CQR but with a multiplicative form.
   - Precisely, relies on RCPS (Bates et al., 2021)

Guarantee:

$$\mathbb{P}\left[\mathbb{E}\left[\text{Average miscoverage on all pixels of a test image}|\text{Cal}\right] \geq \alpha\right] \leq \delta$$

$\to$ Marginal validity on the test, with high probability w.r.t. the calibration set.

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Abstract

Image-to-image regression is an important learning task, used frequently in biological imaging. Current algorithms, however, do not generally offer statistical guarantees that protect against a model's mistakes and hallucinations. To address this, we develop uncertainty quantification techniques with rigorous statistical guarantees for image-to-image regression problems. In particular, we show how to derive uncertainty intervals around each pixel that are guaranteed to contain the true value with a user-specified confidence probability. Our methods work in conjunction

2. Methods

We now formally describe the method for constructing uncertainty intervals. Each pixel in the image will get its own uncertainty interval, as in (1), that is statistically guaranteed to contain the true value with high probability.

How do you understand that?

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Abstract

Image-to-image regression is an important learning task, used frequently in biological imaging. Current algorithms, however, do not generally offer statistical guarantees that protect against a model's mistakes and hallucinations. To address this, we develop uncertainty quantification techniques with rigorous statistical guarantees for image-to-image regression problems. In particular, we show how to derive uncertainty intervals around each pixel that are guaranteed to contain the true value with a user-specified confidence probability. Our methods work in conjunction

- Not a conditional coverage claim!
- The statement is on-average on the test point - easy or hard.

Size-stratified risk. Next, we seek prediction sets that do not systematically make mistakes in difficult parts of the image. Our risk control requirement in Definition 2.1 may be satisfied even if the prediction sets systematically fail to contain the most difficult pixels. For example, if $\alpha = 0.1$ and 90% of pixels are covered by fixed-width intervals of size 0.01, then the requirement is satisfied—however, the sets no longer serve as useful notions of uncertainty. To

2. Methods

We now formally describe the method for constructing uncertainty intervals. Each pixel in the image will get its own uncertainty interval, as in (1), that is statistically guaranteed to contain the true value with high probability.

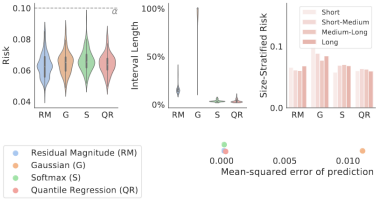
- Hard problem (impossibility results!)
- Introduce metrics to see *if* and *on which underlying regressors* such problem happens.

How do you understand that?

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

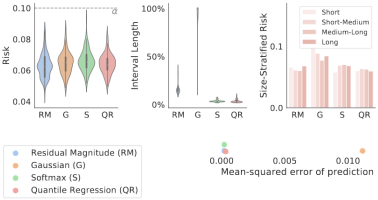
Example of such metrics (see also Feldman et al., 2021) **:**

- Link between the size of the PI and the coverage level $\longrightarrow$



Example of such metrics (see also Feldman et al., 2021) **:**

- Link between the size of the PI and the coverage level $\longrightarrow$
- Localization of the errors $\downarrow$



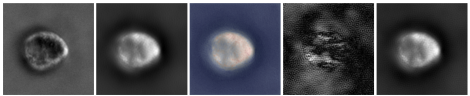


Figure 3. **Examples of quantitative phase reconstructions of leukocytes with uncertainty** shown in the following order: input (we only show one of the two illuminations), prediction, uncertainty visualization (produced with quantile regression), absolute difference between prediction and ground truth (renormalized for visualization), ground truth.

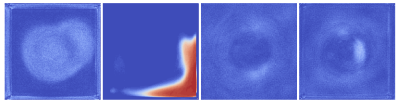


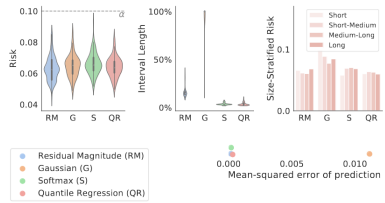
Figure 8. **Spatial variations in miscoverage** in the BSCCM dataset are shown for each of the four methods as a heatmap. Blue represents 0% miscoverage and red represents 100%. The methods are, in order, residual magnitude, gaussian, softmax, and quantile regression.

Figure 2: All images from Angelopoulos et al. (2022)

Image to Image regression with DF-UQ – Angelopoulos et al. (2022)

Example of such metrics (see also Feldman et al., 2021) **:**

- Link between the size of the PI and the coverage level $\longrightarrow$
- Localization of the errors $\downarrow$



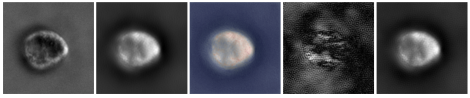


Figure 3. **Examples of quantitative phase reconstructions of leukocytes with uncertainty** shown in the following order: input (we only show one of the two illuminations), prediction, uncertainty visualization (produced with quantile regression), absolute difference between prediction and ground truth (renormalized for visualization), ground truth.

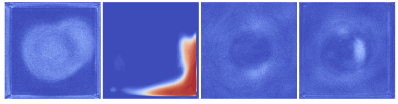


Figure 8. **Spatial variations in miscoverage** in the BSCCM dataset are shown for each of the four methods as a heatmap. Blue represents 0% miscoverage and red represents 100%. The methods are, in order, residual magnitude, gaussian, softmax, and quantile regression.

Figure 2: All images from Angelopoulos et al. (2022)

Take aways:

- Elegant application of SCP with CQR type score
- Test marginal and calibration + train conditional validity guarantees with HP
- Main problem is Test conditionality $\rightarrow$ look at metrics to evaluate which methods performs best!

Where are we now?

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory)

Where are we now?

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)
3. Towards conditional coverage? (practical session) (theory) (case studies)

Where are we now?

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)
3. Towards conditional coverage? (practical session) (theory) (case studies)
4. Beyond exchangeability (methods) (case studies)
5. Computational and statistical trade-offs (methods) (theory)
6. Handling missing data (methods)

Angelopoulos, A. N., Kohli, A. P., Bates, S., Jordan, M., Malik, J., Alshaabi, T., Upadhyayula, S., and Romano, Y. (2022). Image-to-image regression with distribution-free uncertainty quantification and applications in imaging. In *International Conference on Machine Learning*, pages 717–730. PMLR.

Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference: A Journal of the IMA*, 10(2).

Bates, S., Angelopoulos, A., Lei, L., Malik, J., and Jordan, M. (2021). Distribution-free, risk-controlling prediction sets. *Journal of the ACM (JACM)*, 68(6):1–34.

Bian, M. and Barber, R. F. (2023). Training-conditional coverage for distribution-free predictive inference. *Electronic Journal of Statistics*, 17(2):2044 – 2066.

References ii

Chernozhukov, V., Wüthrich, K., and Zhu, Y. (2021). Distributional conformal prediction. *Proceedings of the National Academy of Sciences*, 118(48).

Ding, T., Angelopoulos, A., Bates, S., Jordan, M., and Tibshirani, R. J. (2023). Class-conditional conformal prediction with many classes. In Oh, A., Neumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 64555–64576. Curran Associates, Inc.

Ding, T., Fermanian, J.-B., and Salmon, J. (2026). Conformal prediction for long-tailed classification. In Vondrick, C., Hariharan, B., Raffel, C., Pinto, L., Yang, D., and Faust, A., editors, *International Conference on Learning Representations*, volume 2026, pages 82431–82462.

Feldman, S., Bates, S., and Romano, Y. (2021). Improving Conditional Coverage via Orthogonal Quantile Regression. *arXiv:2106.00394 [cs]*. arXiv: 2106.00394.

Gibbs, I., Cherian, J. J., and Candès, E. J. (2023). Conformal prediction with conditional guarantees. arXiv: 2305.12616.

Guan, L. (2022). Localized conformal prediction: a generalized inference framework for conformal prediction. *Biometrika*, 110(1).

Izbicki, R., Shimizu, G., and Stern, R. B. (2022). CD-split and HPD-split: Efficient conformal regions in high dimensions. *Journal of Machine Learning Research*, 23(87).

Jung, C., Noarov, G., Ramalingam, R., and Roth, A. (2023). Batch multivalid conformal prediction. In *International Conference on Learning Representations*.

Kivaranovic, D., Johnson, K. D., and Leeb, H. (2020). Adaptive, Distribution-Free Prediction Intervals for Deep Networks. In *International Conference on Artificial Intelligence and Statistics*. PMLR.

Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. (2018). Distribution-Free Predictive Inference for Regression. *Journal of the American Statistical Association*.

Lei, J. and Wasserman, L. (2014). Distribution-free prediction bands for non-parametric regression. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 76(1).

Romano, Y., Barber, R. F., Sabatti, C., and Candès, E. (2020). With Malice Toward None: Assessing Uncertainty via Equalized Coverage. *Harvard Data Science Review*, 2(2).

Romano, Y., Patterson, E., and Candès, E. (2019). Conformalized Quantile Regression. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Sesia, M. and Romano, Y. (2021). Conformal prediction using conditional histograms. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Vovk, V. (2012). Conditional Validity of Inductive Conformal Predictors. In *Asian Conference on Machine Learning*. PMLR.

Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer US.

Zaffran, M., Dieuleveut, A., Josse, J., and Romano, Y. (2024). Predictive uncertainty quantification with missing values. Preprint submitted to *Journal of Machine Learning Research*, arXiv arXiv:2405.15641.

CC BY NC SA